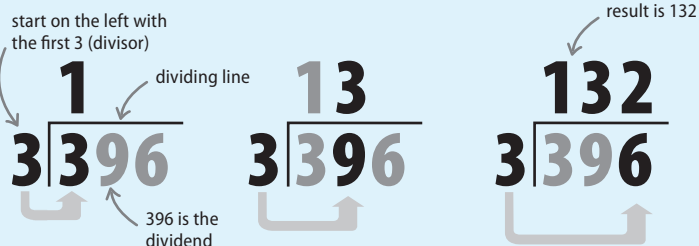


Short division

Short division is used to divide one number (the dividend) by another whole number (the divisor) that is less than 10.



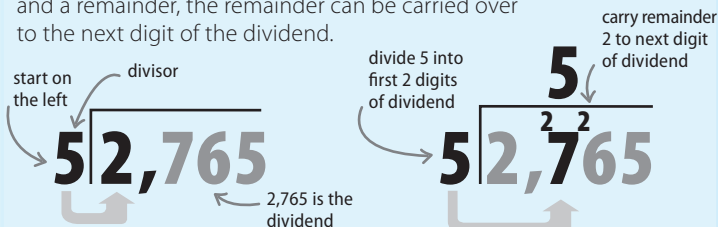
Divide the first 3 into 3. It fits once exactly, so put a 1 above the dividing line, directly above the 3 of the dividend.

Move to the next column and divide 3 into 9. It fits three times exactly, so put a 3 directly above the 9 of the dividend.

Divide 3 into 6, the last digit of the dividend. It goes twice exactly, so put a 2 directly above the 6 of the dividend.

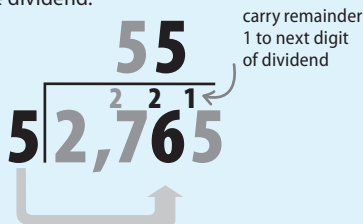
Carrying numbers

When the result of a division gives a whole number and a remainder, the remainder can be carried over to the next digit of the dividend.



Start with number 5. It does not divide into 2 because it is larger than 2. Instead, 5 will need to be divided into the first two digits of the dividend.

Divide 5 into 27. The result is 5 with a remainder of 2. Put 5 directly above the 7 and carry the remainder.



Divide 5 into 26. The result is 5 with a remainder of 1. Put 5 directly above the 6 and carry the remainder 1 to the next digit of the dividend.

Divide 5 into 15. It fits three times exactly, so put 3 above the dividing line, directly above the final 5 of the dividend.

LOOKING CLOSER

Converting remainders

When one number will not divide exactly into another, the answer has a remainder. Remainders can be converted into decimals, as shown below.

$$\begin{array}{r} \text{remainder} \\ 22 \text{ r } 2 \\ 4 \overline{) 90} \end{array}$$

$$\begin{array}{r} 22. \\ 4 \overline{) 90.0} \end{array}$$

Remove the remainder, 2 in this case, leaving 22. Add a decimal point above and below the dividing line. Next, add a zero to the dividend after the decimal point.

$$\begin{array}{r} 22. \\ 4 \overline{) 90.0} \end{array}$$

Carry the remainder (2) from above the dividing line to below the line and put it in front of the new zero.

$$\begin{array}{r} 22.5 \\ 4 \overline{) 90.0} \end{array}$$

Divide 4 into 20. It goes 5 times exactly, so put a 5 directly above the zero of the dividend and after the decimal point.

LOOKING CLOSER

Making division simpler

To make a division easier, sometimes the divisor can be split into factors. This means that a number of simpler divisions can be done.

$$816 \div 6 = 408 \quad \text{divisor is 6, which is } 2 \times 3. \text{ Splitting 6 into 2 and 3 simplifies the sum}$$

$$816 \div 2 = 408 \quad \text{divide by first factor of divisor}$$

$$408 \div 3 = 136 \quad \text{divide by second factor of divisor}$$

result is 136

This method of splitting the divisor into factors can also be used for more difficult divisions.

$$405 \div 15 = 27 \quad \text{splitting 15 into 5 and 3, which multiply to make 15, simplifies the problem}$$

$$405 \div 5 = 81 \quad \text{divide by first factor of divisor}$$

$$81 \div 3 = 27 \quad \text{divide result by second factor of divisor}$$

result is 27